Math Kangaroo Practice

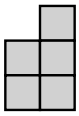
Grade 3-4 · 20 questions · harder · seed 33009

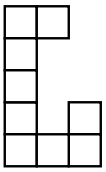
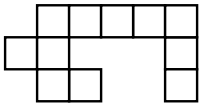
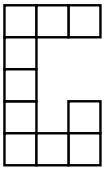
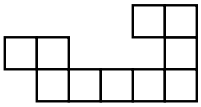
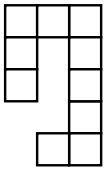
Part A: Each question is worth 3 points

1. Tom, who is older than Ben by 1 year and 2 days, was born on December 12, 2007. When was Ben born?

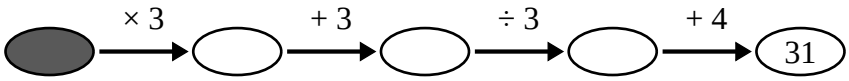
A. December 10, 2006 ***B.*** December 14, 2006 ***C.*** December 14, 2009
D. December 12, 2008 ***E.*** December 14, 2008

2. Which of the five pieces fits together with the grey piece to make a rectangle? Pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

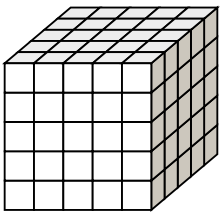
3. Tom wrote a favourite number in the dark cloud and did the calculations shown by the arrows, one after the other. The result was 31.



What is Tom's favourite number?

A. 25 ***B.*** 26 ***C.*** 27 ***D.*** 31 ***E.*** 36

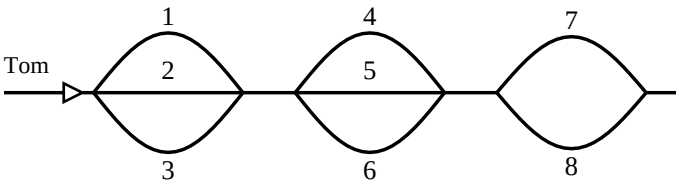
4. A cube with sides of 5 cm is painted grey all over and then cut into small cubes with sides of 1 cm.



How many of the small cubes have exactly 2 painted faces?

A. 18 ***B.*** 24 ***C.*** 36 ***D.*** 54 ***E.*** 60

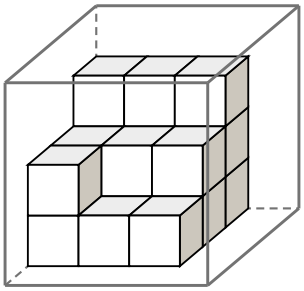
5. Tom walks along the road from left to right and writes down every number on the way.



How many different groups of numbers can Tom write?

A. 6 ***B.*** 18 ***C.*** 20 ***D.*** 22 ***E.*** 23

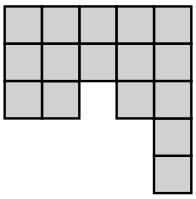
6. Daniella has a glass aquarium shaped like a cube, 4 dm long, 4 dm wide and 4 dm tall. Daniella has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.



At most how many more such cubes can Daniella put into the aquarium?

A. 64 ***B.*** 51 ***C.*** 46 ***D.*** 45 ***E.*** 19

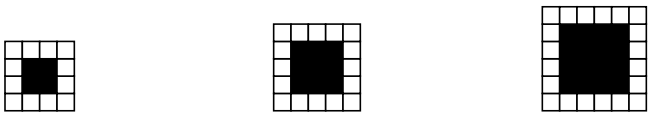
7. Tom wants to add one more piece to the grey piece to make the smallest possible rectangle.



How many little squares must that piece have?

A. 25 ***B.*** 20 ***C.*** 14 ***D.*** 9 ***E.*** 8

8. These diagrams follow a pattern: a black square with a white border around it. How many white cells will the diagram after the next one have?



12 white cells 16 white cells 20 white cells

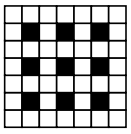
A. 32 ***B.*** 28 ***C.*** 27 ***D.*** 24 ***E.*** 23

Part B: Each question is worth 4 points

9. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many black cells will the 8th diagram have?

8 white cells

21 white cells



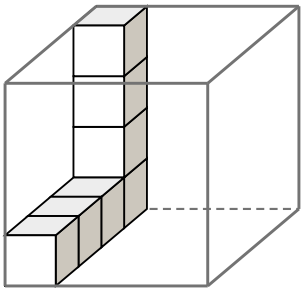
40 white cells

A. 16 ***B.*** 49 ***C.*** 64 ***D.*** 65 ***E.*** 81

10. What is the smallest number whose digits add up to 28?

A. 9991 ***B.*** 2989 ***C.*** 2899 ***D.*** 2000 ***E.*** 1999

11. Sara's box is 4 cubes long, 4 cubes deep and 4 cubes tall. Some cubes are already inside, as in the picture. Every cube stands on the bottom or on another cube.



How many more cubes fit into the box?

A. 9 ***B.*** 46 ***C.*** 51 ***D.*** 56 ***E.*** 57

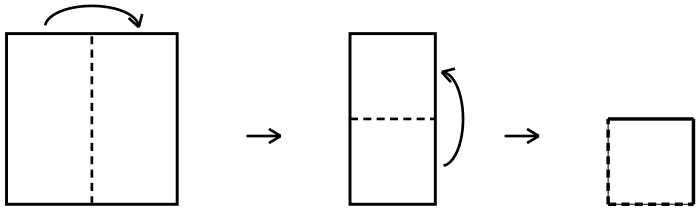
12. Which word has exactly 1 different letter in common with PROBLEM?

A. BANANA ***B.*** GARDEN ***C.*** TRIANGLE ***D.*** BUTTERFLY ***E.*** NUMBER

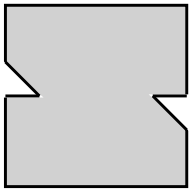
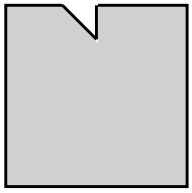
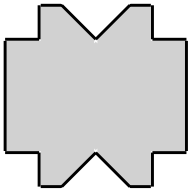
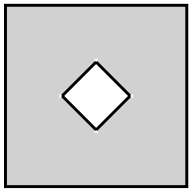
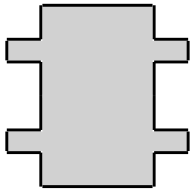
13. Only the even numbers from 2 to 120 are written down: 2, 4, 6, ..., 120.
How many digits are written?

A. 138 ***B.*** 132 ***C.*** 129 ***D.*** 128 ***E.*** 127

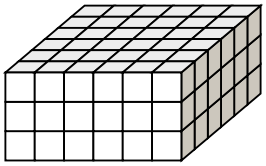
14. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square is cut off, and the paper is unfolded.



Which design **can** be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

15. A brick 6 cubes long, 6 cubes deep and 3 cubes tall is made of small cubes. It is painted on the outside, also underneath, and then taken apart.



How many small cubes have exactly 2 painted faces?

A. 8 ***B.*** 32 ***C.*** 36 ***D.*** 40 ***E.*** 48

16. Basil was born on May 25, 2013 and Peter on June 21, 2013.
By how many days is Basil older than Peter?

A. 27 ***B.*** 28 ***C.*** 29 ***D.*** 31 ***E.*** 32

17. Nina thinks of a number, divides by 5, then multiplies by 4, then takes away 1, and gets 19.
Which number did Nina think of?

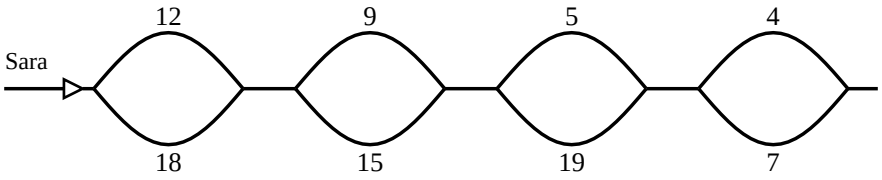
A. 19 ***B.*** 25 ***C.*** 26 ***D.*** 27 ***E.*** 30

Part C: Each question is worth 5 points

18. Leo types the numbers from 1 to 150 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,150.
How many keys does Leo press (each digit and each comma is one key)?

A. 300 ***B.*** 490 ***C.*** 491 ***D.*** 492 ***E.*** 500

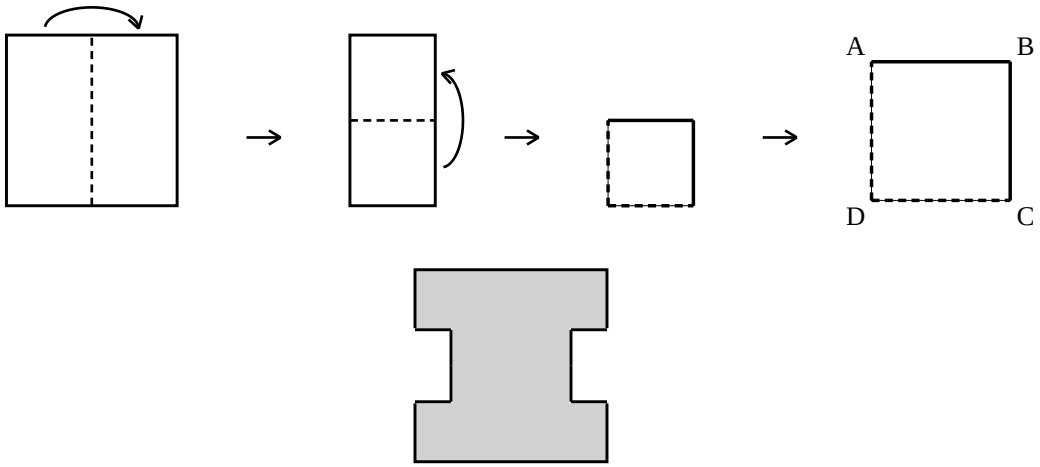
19. Sara walks along the road from left to right and adds up the numbers along the way. The total is 45.



Which number did Sara see at the second fork?

A. 5 ***B.*** 7 ***C.*** 12 ***D.*** 15 ***E.*** 19

20. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square was cut off, and after unfolding the paper looks like the bottom picture.



Which corner was cut off?

A. corner A ***B.*** corner B ***C.*** corner C ***D.*** corner D ***E.*** it cannot be made with one cut

Answer Key

1.	**E**	**2.**	**C**	**3.**	**B**	**4.**	**C**	**5.**	**B**	**6.**	**D**
7.	**D**	**8.**	**B**	**9.**	**C**	**10.**	**E**	**11.**	**E**	**12.**	**A**
13.	**E**	**14.**	**D**	**15.**	**C**	**16.**	**A**	**17.**	**B**	**18.**	**C**
19.	**D**	**20.**	**C**								

Solutions

1. E (December 14, 2008) — Older means born earlier, so Ben was born 1 year and 2 days **after** December 12, 2007: 1 year later is December 12, 2008, then 2 more days: December 14, 2008. [set-03/by_level/Grade_03-04/2007 Q12 · younger_birthday]

2. C — The grey piece and the right piece together fill a 3 × 5 rectangle: 15 – 5 = 10 squares, in exactly the shape of the gap. Piece C (turned) fills it; the others have a square too many, too few, or in the wrong place. [set-03/by_level/Grade_03-04/2007 Q8 · complete_rectangle]

3. B (26) — Go backwards and do the opposite: 31 – 4 = 27, 27 × 3 = 81, 81 – 3 = 78, 78 ÷ 3 = 26. [set-03/by_level/Grade_03-04/2007 Q5 · favourite_number]

4. C (36) — They sit on the edges, but not at the corners. 12 edges with 5 – 2 = 3 such cubes each: 12 × 3 = 36. [set-03/by_level/Grade_03-04/2007 Q17 · two_faces]

5. B (18) — The forks have 3, 3, 2 branches; every choice can go with every other: 3 × 3 × 2 = 18. [set-03/by_level/Grade_03-04/2007 Q1 · how_many_routes]

6. D (45) — The aquarium holds 4 × 4 × 4 = 64 cubes. There are already 19 inside (count column by column, also the hidden ones). 64 – 19 = 45. [set-03/by_level/Grade_03-04/2007 Q14 · how_many_more]

7. D (9) — The smallest rectangle around the grey piece is 5 × 5 = 25 squares. 25 – 16 = 9. [set-03/by_level/Grade_03-04/2007 Q8 · squares_missing]

8. B (28) — The border of an 8 × 8 square has 8 + 8 + 6 + 6 = 28 cells (or 8 × 8 – 6 × 6 = 28). [set-03/by_level/Grade_03-04/2007 Q21 · border_next]

9. C (64) — The 8th diagram is 17 × 17 = 289 cells, with 8 × 8 = 64 black ones. [set-03/by_level/Grade_03-04/2007 Q21 · black_and_total]

10. E (1999) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 1999: 1 + 9 + 9 + 9 = 28. [set-03/by_level/Grade_03-04/2007 Q4 · smallest_with_sum]

11. E (57) — The box holds 4 × 4 × 4 = 64 cubes. 7 are inside, so 64 – 7 = 57 more. [set-03/by_level/Grade_03-04/2007 Q14 · long_box]

12. A (BANANA) — BANANA shares B with PROBLEM. GARDEN shares 2; BUTTERFLY shares 4; NUMBER shares 4; TRIANGLE shares 3. [set-03/by_level/Grade_03-04/2007 Q3 · which_word]

13. E (127) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 11. 4 × 1 + 45 × 2 + 11 × 3 = 127. [set-03/by_level/Grade_03-04/2007 Q23 · even_only]

14. D — A single cut appears four times, the same on the left and right and on the top and bottom. Only design D looks like that (a hole in the middle). [set-03/by_level/Grade_03-04/2007 Q24 · which_can]

15. C (36) — On each edge, the cubes between the two corners: the brick has 4 edges of each length 6, 6, 3, so 4 × (4 + 4 + 1) = 36. [set-03/by_level/Grade_03-04/2007 Q17 · painted_brick]

16. A (27) — May has 31 days, so from May 25 to the end of the month is 6 days, then 21 more days to June 21, 2013: 27 days. [set-03/by_level/Grade_03-04/2007 Q12 · days_between]

17. B (25) — Go backwards: 19 + 1 = 20, 20 ÷ 4 = 5, 5 × 5 = 25. [set-03/by_level/Grade_03-04/2007 Q5 · think_of_a_number]

18. C (491) — Digits: 9 × 1 + 90 × 2 + 51 × 3 = 342. Commas: one fewer than the numbers, 149. 342 + 149 = 491. [set-03/by_level/Grade_03-04/2007 Q23 · with_commas]

19. D (15) — Only one route gives 45: 18 + 15 + 5 + 7 = 45. At the second fork: 15. [set-03/by_level/Grade_03-04/2007 Q1 · which_branch]

20. C (corner C) — The design shows notches in the middle of the left and right sides. That comes from cutting corner C (where one fold meets a paper edge). [set-03/by_level/Grade_03-04/2007 Q24 · which_corner]